STRONG SPLITTING WITH A SCHEME SOURCE WHEN S, T ARE FLAT AND FINITELY PRESENTED

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers within this span. The section-variant input is Omitted+Hint in the source and is declared external, never proven here.

Allowed standard prerequisites (editorial).

- Strong splitting existence. lemma-strong-splitting (carded L26658). The construction of U' whose scheme nature is now shown.
- Scheme nature of the finite part with a section. lemma-finite-separated-flat-locally-finite-presentation (carded L26657) and lemma-finite-plus-section (same chapter, NOT carded, declared external). U' = (R_s/U, e)_fin is a scheme separated over U.

Source reference. Original lemma 04RR: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2168–2168. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04RI: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2082–2082. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03FM: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2514–2514. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04RL: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2463–2463. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 04RL assume in addition that s, t are flat and locally of finite presentation. Then there exists a scheme U', a separated étale morphism $U' \to U$, and a point $u' \in U'$ lying over u with $\kappa(u) = \kappa(u')$ such that the restriction $R' = R|_{U'}$ of R to U' is strongly split over u'.*

Date: stacks-project:spaces-more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. This follows from the construction of U' in the proof of Lemma 03FM because in this case $U' = (R_s/U, e)_{fin}$ is a scheme separated over U by Lemmas 04RI and 04RR. □